DSAI LAB PROJECT · MILESTONE 1 REVIEW

Detecting AI-generated voices in Indian languages

Problem definition and literature review

Team · Group-3

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01 · THE PROBLEM

AI can now copy anyone’s voice in seconds

Easy to make

Free tools like XTTS-v2 can clone a voice from about 6 seconds of audio.

Hard to hear

A cloned voice keeps the person’s accent and tone. Most people cannot tell by ear.

Used for scams

Fraudsters pretend to be family or bank officials, in Hindi, Tamil, Telugu and English.

But most detectors are trained and tested only on English speech.

Our question

> Can we tell whether a speech clip is spoken by a real person or made by AI, in English, Hindi, Tamil and Telugu, even for languages and voice tools the model has never seen?

OUR GOAL

Real voice or AI voice?

NOT OUR GOAL

Is this call a scam?

Scams are why this matters. The model only judges the voice, not what is being said.

03 · SCOPE

What we will and won’t do

In scope

- Short speech clips: real or AI-generated
- English, Hindi, Tamil and Telugu
- New fake clips made with open TTS tools
- Fine-tuning a pretrained speech model (XLS-R)
- Tests on new languages, new voice tools and phone audio
- Web app and Telegram bot demo

Out of scope

- Scam or intent detection
- Who is speaking (speaker verification)
- Blocking live phone calls
- Other languages
- Video deepfakes
- Full production rollout

04 · STAKEHOLDERS

Who this helps



General public

The main targets of cloned-voice calls and voice notes. They risk losing money and feel real distress.



Elderly users

Most at risk from “family emergency” calls. They need a simple way to check a voice note.



Banks

Lose money to fraud, and use voice checks for customers that cloning can fool.

05 · OBJECTIVES

Six goals, each with a number

#	Goal	How we measure it
1	Build the dataset	2,000+ labelled fake clips per language, from 3 TTS models
2	Fine-tune XLS-R	EER ≤ 2% on ASVspoof 2019; ≤ 5% on our Indian-language test
3	Beat the baselines	Lower EER than the CNN and AASIST-L on every test
4	Test generalisation	New language, new voice tool, real-world and phone audio: EER below 15%
5	Explain decisions	Show which seconds of the clip drove each prediction
6	Working demo	Result in under 5 s for a 10 s clip; web app and Telegram bot

How detectors have improved

Approach	Example	Best EER	Main weakness
Hand-made features	LFCC-GMM	8.09%	Misses modern AI voices
CNN on spectrograms	LCNN	1.92%	Learns one English dataset
Raw-audio models	AASIST	0.83%	Breaks on new data
Self-supervised models	XLS-R + AASIST	0.82%*	Not tested on Indian languages

EER on ASVspoof 2019 LA (English). Lower is better. *On the harder ASVspoof 2021 LA set.

07 · THE CATCH

Lab results don’t hold up in the real world

1.2%

error on its own lab test (ASVspoof 2019)

→

37.2%

error on real-world fakes from the internet (In-the-Wild)

Same model (RawGAT-ST), about 30 times worse. The newest benchmark, ASVspoof 5, shows 29–36% for standard baselines. **Every one of these benchmarks is English only.**

Equal Error Rate (EER)

The model gives every clip a “fake score”. We choose the cut-off where two kinds of mistake happen equally often:

Fake passed as real

An AI voice the model missed

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Real flagged as fake

A human voice wrongly accused

That shared error rate is the EER. **Lower is better.** It is the standard metric in this field. We also report ROC-AUC, plus accuracy and F1 for the demo.

09 · DATA

Free datasets we will use

Dataset	Language	What it has	Our use
ASVspoof 2019 LA	English	Real + fake speech	Train and compare
In-the-Wild	English	Real-world fakes, 37.9 h	Test only
Kathbath	12 Indian	Real speech, 1,684 h	Real Hindi, Tamil, Telugu
IndicSynth	12 Indian	Fake speech, 4,000 h	Main Indian fakes
Our own set	Hindi, Tamil, Telugu	2,000+ fakes per language, 3 TTS models	New-voice-tool tests

Our fakes come from Indic Parler-TTS, IndicF5 and MMS-TTS, run on Kaggle’s free GPU. Total cost: zero.

What’s missing, and our answer

Gap in existing work	What we do
Benchmarks are English only	Test on Hindi, Tamil, Telugu and English
Fail on new data (29–37% EER)	Tests on unseen languages and voice tools
New Indic voice tools not tested	Our fakes from Indic Parler-TTS, IndicF5, MMS-TTS
No phone-audio testing	Train and test with phone-quality audio
Only a score, no reason	Show which seconds look fake
Not available to the public	Free web app and Telegram bot

How our system compares

Feature	GMM	CNN / AASIST	XLS-R based	Ours
Learns its own features	No	Yes	Yes	Yes
Multilingual pretraining	No	No	Yes	Yes
Tested on Indian languages	No	No	Limited	Yes
Unseen language / tool tests	No	Limited	Limited	Yes
Phone-audio tests	No	No	Limited	Yes
Explains its decision	No	No	No	Yes
Public demo	No	No	No	Yes